

Development of Predictive Economic Decision-Support Frameworks for Modernizing U.S. Trade and Industrial Policy and Strengthening Domestic Manufacturing Resilience and Competitiveness

A Research Paper

Olusoji Adenekan

[Author Credentials / Field of Expertise]

[Affiliation, if applicable]

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ABSTRACT

The United States' extensive federal trade, manufacturing, and economic data infrastructure remains limited by fragmentation across agencies, insufficient industry-level granularity, lagging indicators, weak lower-tier supply-chain visibility, and a general orientation toward retrospective description rather than forward-looking, predictive analysis. These limitations constrain the capacity of policymakers, economic-development organizations, and manufacturers to anticipate how trade restrictions, foreign supply disruptions, and other shocks propagate through interconnected U.S. industries. This paper develops a Predictive Trade and Industrial Intelligence Framework that integrates trade, manufacturing, supply-chain, investment, labor, and energy data; applies hybrid econometric and machine-learning forecasting, network-based supply-chain risk modeling, and comparative policy-scenario simulation; and translates the results into standardized indicators and decision-support dashboards.

The framework's core econometric and predictive components were tested empirically on a real panel dataset covering 19 U.S. manufacturing subsectors from 2015 through 2025 (N = 209 industry-year observations), assembled from the U.S. Bureau of Economic Analysis, the U.S. Bureau of Labor Statistics, and the U.S. International Trade Commission. A two-way fixed-effects panel regression found trade activity positively associated with industry output, though not statistically significant at the 5 percent level under a wild-cluster bootstrap appropriate to this panel's modest number of cross-sectional clusters. A directly computed Import-Origin Concentration Index has a consistently negative estimated coefficient across specifications; it is statistically significant in several less-controlled specifications but not in the preferred two-way fixed-effects specification or in a trade-by-concentration interaction specification testing for differential subsector vulnerability. A pseudo-out-of-sample, rolling-origin, one-year-ahead forecasting exercise using only lagged predictors found that a naive persistence benchmark was difficult for econometric and machine-learning models to outperform.

The study's principal contribution is the applied integration of fragmented federal data sources into a single analytical panel, combined with panel econometric modeling, forecast evaluation, and an open decision-support implementation — and the study integrates these established methods within a unified applied framework for U.S. trade and manufacturing analysis. The study constitutes preliminary evidence toward the first three phases of a proposed five-phase national implementation, while identifying specific, disclosed gaps — dependency-network mapping, causal scenario analysis, and retrospective disruption validation — that remain for future work.

Keywords: trade policy, industrial resilience, supply-chain risk, predictive analytics, econometric forecasting, scenario simulation, manufacturing competitiveness, decision-support systems.

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CHAPTER ONE: INTRODUCTION

1.1 Background and Context

The United States economy is embedded in a global network of suppliers, manufacturers, distributors, and end markets whose interdependencies have become increasingly consequential to national economic security. Trade liberalization, offshoring, and the geographic concentration of critical manufacturing capacity have combined to produce supply chains that are efficient under stable conditions but fragile under stress. Disruptions originating far outside U.S. borders — a factory closure abroad, an export restriction on a critical mineral, a shipping bottleneck, a geopolitical conflict — can propagate rapidly through supplier tiers to disrupt domestic production, employment, and prices.

Recent years have demonstrated this repeatedly: semiconductor disruptions constrained automotive and electronics production; critical-mineral and rare-earth restrictions exposed dependencies in clean-energy, defense, and advanced-manufacturing industries; pharmaceutical ingredient shortages revealed similar dependencies in health. Each episode exposed the same weakness: policymakers generally lacked the granular, forward-looking, interconnected view of supply chains needed to anticipate disruptions rather than react to them.

The federal statistical apparatus — Census, BEA, BLS, USITC, USTR, Commerce, the Federal Reserve — produces an enormous volume of trade, manufacturing, investment, labor, and macroeconomic data, a genuine national asset. But it was largely designed for retrospective description rather than predictive, supply-chain-aware decision support: siloed across agencies, published on lagging schedules, aggregated above the granularity needed to trace lower-tier relationships, and not integrated into models capable of forecasting how a shock propagates.

This paper responds to that gap. It develops and documents a Predictive Trade and Industrial Intelligence Framework intended to integrate fragmented federal and industry data, apply econometric and predictive modeling, simulate policy and disruption scenarios, and translate results into decision-support tools for policymakers, economic-development agencies, and manufacturers — designed to be national in scope, modular across industries and regions, and continuously updated.

1.2 Statement of the Problem

Several structural limitations reduce federal data's usefulness for anticipatory decision-making. Information is fragmented across agencies not designed to interoperate. Published statistics are often too coarse to reveal specific inputs, suppliers, or components. Many indicators are lagging by design. Lower-tier supplier visibility — the second-, third-, and fourth-tier firms where

disruptions frequently originate — is limited. And existing tools generally describe past conditions rather than predicting how a prospective shock would propagate.

The consequence: trade policy, resilience investment, and competitiveness strategy are frequently formulated with an incomplete, retrospective picture. Policymakers may be unable to anticipate exposure to a prospective disruption; manufacturers may be unable to quantify supplier concentration; economic-development agencies may lack evidence to prioritize investment where it matters most.

There is therefore a documented, unmet need for a scalable analytical framework integrating diverse economic data, applying predictive and scenario-based modeling, and translating results into actionable insight for the stakeholders responsible for U.S. trade policy, manufacturing resilience, and industrial competitiveness.

1.3 Purpose of the Study

The purpose of this study is to design, document, and validate a predictive economic decision-support framework capable of integrating trade, manufacturing, supply-chain, investment, labor, and energy data; forecasting trade flows, production, and manufacturing conditions; identifying supply-chain concentration and dependency risk; simulating the economic effects of disruptions and policy interventions; and presenting results through tools usable by policymakers, economic-development organizations, and manufacturers — with a replicable methodology extensible beyond the initial pilot scope.

1.4 Research Objectives

This study pursues the following objectives:

- To assess the current state of U.S. trade, manufacturing, supply-chain, investment, labor, and energy data infrastructure and identify specific gaps in granularity, timeliness, interoperability, and lower-tier supply-chain visibility.
- To map the relationships among foreign suppliers, imported inputs, domestic manufacturers, and downstream industries in order to identify critical dependencies and potential bottlenecks.
- To develop econometric, time-series, and machine-learning models capable of forecasting trade flows, domestic production, investment, import dependency, and manufacturing conditions.
- To develop supply-chain risk models capable of quantifying supplier concentration, critical-input dependency, and domestic capacity gaps.

- To develop scenario-simulation and stress-testing models capable of evaluating the effects of tariffs, foreign export restrictions, geopolitical disruptions, shortages, and demand shocks, as well as alternative trade and industrial-policy interventions.
- To design and validate decision-support dashboards and indicators that translate model outputs into a form usable by policymakers, economic-development agencies, and manufacturers.
- To establish a replicable, modular methodology that allows the framework to be scaled across additional industries and regions.

1.5 Research Questions

The study is organized around the following research questions:

- RQ1: What are the principal gaps in existing federal and industry data infrastructure that limit predictive, supply-chain-aware analysis of U.S. trade and manufacturing conditions?
- RQ2: How can trade, manufacturing, supply-chain, investment, labor, and energy data be integrated into a coherent framework capable of forecasting industrial and trade conditions?
- RQ3: To what extent can econometric and machine-learning models forecast trade flows, production, and import dependency with sufficient accuracy to support policy and investment decisions?
- RQ4: How can supply-chain risk — including supplier concentration and critical-input dependency — be quantified in a way that is comparable across industries?
- RQ5: How do simulated trade disruptions, foreign export restrictions, and alternative policy interventions affect production, prices, employment, investment, and downstream industries within the framework?
- RQ6: What decision-support formats are most effective at translating model output into actionable guidance for policymakers, economic-development organizations, and manufacturers?

1.6 Significance of the Study

This study's significance spans three dimensions: economic security, industrial competitiveness, and evidence-based policymaking. Economically, the framework addresses a documented vulnerability in the U.S. capacity to anticipate, rather than merely respond to, trade and manufacturing disruptions, supporting earlier identification of critical foreign dependencies and concentration.

For competitiveness, the framework offers manufacturers, regional economic-development organizations, and industry associations a shared, standardized indicator system that can inform investment, reshoring, and diversification decisions with rigor most small and mid-sized manufacturers cannot develop independently.

For policy, the framework is designed as a decision-support tool, not a black box. In its fully developed and causally validated form, its scenario-simulation components are intended to let policymakers compare projected effects of alternative interventions — tariffs versus domestic-capacity investment versus supplier diversification — before enactment. The pilot-scale tool demonstrated here (Section 4.6) is an exploratory, non-causal illustration of that intended capability, not yet the decision-grade causal simulation this longer-term goal describes.

Taken together, these contributions support a national interest in strengthening the analytical infrastructure underlying U.S. trade and industrial policy, with outputs — an open, replicable methodology, publications, and technical documentation — intended to benefit the field beyond the scope of any single project.

1.7 Scope and Delimitations

This study is delimited to developing and validating a decision-support framework; it does not itself recommend any specific tariff, sanction, or regulatory action. The framework is industry-agnostic in structure and validated, in Chapter Four, on 19 real NAICS-3 manufacturing subsectors spanning a wide range of supply-chain structures — from Computer and Electronic Products to Wood Products and Textile Mills — selected as the full set of subsectors for which all four federal sources (BEA, BLS Employment, BLS Productivity, USITC) provide comparable annual detail over 2015–2025.

The study relies on publicly available federal datasets and established economic-modeling methodologies. Its predictive models are subject to the general limitations of econometric and machine-learning forecasting — sensitivity to data quality, structural breaks, and unmodeled exogenous shocks — addressed in the uncertainty-analysis and human-in-the-loop components of the methodology and discussed further in Chapter Five.

1.8 Definition of Key Terms

- **Industrial Resilience and Competitiveness Indicators:** A standardized set of quantitative measures reflecting an industry's exposure to supply disruption, domestic production capacity, and relative competitive position.
- **Supply-Chain Concentration:** The degree to which a critical input or component is sourced from a small number of suppliers, countries, or regions, increasing vulnerability to disruption.

- Lower-Tier Visibility: The extent to which second-, third-, and fourth-tier suppliers — those beyond direct, tier-one relationships — can be identified and monitored.
- Scenario Simulation: A modeling technique used to estimate the economic effects of a hypothetical event, such as a tariff, export restriction, or geopolitical disruption, under specified assumptions.
- Stress Testing: The systematic evaluation of a system's performance under adverse or extreme conditions, adapted here from financial-sector practice to trade and manufacturing systems.
- Decision-Support Dashboard: An interactive analytical tool that presents model outputs, indicators, and scenario results in a form designed for use by non-technical policy and industry stakeholders.
- Human-in-the-Loop Validation: A model-governance practice in which subject-matter experts review, interpret, and, where necessary, override or adjust model outputs before they inform decisions.

1.9 Organization of the Paper

Chapter One has introduced the problem, objectives, questions, and significance. Chapter Two reviews the relevant literature and situates the framework within it. Chapter Three describes the five-phase methodology, distinguishing the full proposed architecture from the narrower empirical specification estimated here. Chapter Four presents pilot-scale empirical findings from real federal data across 19 manufacturing subsectors, 2015–2025 — genuine empirical estimates, not simulated results, though pilot-scale findings rather than a comprehensive national deployment. Chapter Five discusses implications, contributions, limitations, and future research.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter reviews literature in four areas underlying the proposed framework: U.S. trade and industrial data infrastructure; supply-chain visibility and vulnerability; predictive and econometric modeling; and scenario-based policy simulation. It concludes by identifying specific gaps this study addresses and presenting the conceptual framework guiding the methodology.

2.2 U.S. Trade Policy and Economic Data Infrastructure

The federal government maintains a wide array of trade and industrial datasets — Census trade statistics, BEA input-output accounts, BLS employment and price indices, USITC trade and tariff data, Federal Reserve production indices — representing substantial public investment in economic measurement.

A recurring theme is the tension between data breadth and usability for real-time, cross-agency analysis. Classification systems are generally too aggregated to reveal firm-level dependencies; publication lags appropriate for macro surveillance limit usefulness for anticipatory risk management; and methodological differences across agencies complicate integration into a single pipeline. These limitations are consistent with GAO's own finding that federal agencies coordinating on supply-chain resilience face persistent barriers to data collection and cross-agency sharing (U.S. Government Accountability Office, 2023), motivating this framework's data-integration objectives.

2.3 Supply Chain Visibility and Vulnerability Analysis

Research accelerated by disruptions in semiconductors, critical minerals, and pharmaceutical inputs has examined supply-chain mapping and vulnerability. Network-based approaches represent supply chains as directed graphs, identifying concentration risk and propagation paths; Acemoglu, Carvalho, Ozdaglar, and Tahbaz-Salehi (2012) demonstrate that microeconomic, sector-level shocks can generate aggregate fluctuations through exactly these input-output linkages, motivating this study's interest in concentration as a propagation channel. Antràs and Chor's (2013) analysis of global value chain ownership and control provides conceptual grounding for treating import-origin structure as informative about upstream exposure, though this study's country-level HHI (Section 4.3) cannot observe the firm-level relationships that literature examines directly.

Lower-tier visibility is a persistent, largely unresolved challenge: tier-one relationships are observable with reasonable accuracy, but visibility deteriorates sharply at the tiers where bottlenecks most frequently originate. This study draws on both network-mapping and

inference-based approaches within the constraints of public data. Separately, Autor, Dorn, and Hanson's (2013) analysis of Chinese import competition remains a central reference for import competition's measurable effects on domestic manufacturing, motivating this study's interest in import activity and output (Chapter Four) — though their instrumented causal identification strategy is considerably stronger than the associational panel regression this study estimates with 19 industry-level observations.

2.4 Predictive Analytics and Econometric Forecasting in Trade and Manufacturing

Econometric forecasting of trade and manufacturing output has a long tradition (VAR, error-correction, structural gravity models). Machine-learning methods — gradient-boosted trees, random forests, neural architectures — have more recently shown improved accuracy over linear models when data are sufficient, at some cost to interpretability. Mullainathan and Spiess (2017) frame machine learning as addressing prediction rather than parameter identification, cautioning that predictive accuracy does not provide the causal estimates many policy questions require — a distinction this study maintains explicitly (Section 4.4) in labeling its trade coefficients associational, not causal.

A recurring finding is that hybrid econometric/ML approaches tend to outperform either alone, particularly around structural breaks. This informs the hybrid strategy adopted here (Chapter Three), though this study's own results (Section 4.5) do not reproduce that pattern: at this panel's scale, naive persistence outperformed both linear and machine-learning approaches, illustrating that the literature's general pattern does not automatically transfer to every application.

2.5 Scenario Analysis, Stress Testing, and Policy Simulation

Scenario analysis and stress testing, established in financial-sector risk management, have increasingly been adapted to trade and industrial policy. Caldara, Iacoviello, Molligo, Prestipino, and Raffo (2020) find that trade-policy uncertainty itself reduces investment even before any tariff takes effect — directly relevant to this framework's goal of evaluating prospective actions before enactment. CGE models capture economy-wide tariff and trade-agreement effects that partial-equilibrium approaches omit; agent-based and network-simulation models complement CGE by capturing firm-level responses.

The literature emphasizes comparing multiple candidate interventions under a consistent framework rather than assessing each in isolation — incorporated directly into this framework's Phase 2 and Phase 3 methodology.

2.6 Industrial Resilience and Competitiveness Indicators

A growing literature proposes composite indicators for industrial resilience, drawing on supplier concentration, import dependency, capacity utilization, and workforce and capital trends. Consensus favors indicators that are transparent, comparable, and recurring rather than one-time — principles informing this framework's Phase 1 indicators.

This sits within a broader resurgence of industrial-policy research. Juhász, Lane, and Rodrik (2024) find that better-identified recent studies show more positive industrial-policy effects than an older, more skeptical literature, though success depends heavily on institutional design and monitoring. This framework is positioned to supply the firm- and industry-level evidence base that literature identifies as necessary for well-designed policy, not to advocate for industrial policy as such.

2.7 Gaps in Existing Literature and Tools

Methods exist individually for data integration, supply-chain mapping, forecasting, and scenario simulation, but among the literature and public federal tools reviewed for this study, integrated frameworks combining all four into one continuously updated system applicable across U.S. manufacturing appear limited. Most existing tools are narrowly scoped, built on inaccessible proprietary data, or retrospective rather than forward-looking.

This study addresses that gap directly, combining data-integration methodology, supply-chain mapping, hybrid forecasting, and comparative scenario simulation into one modular framework, validated through a phased methodology designed for national scaling.

2.8 Conceptual Framework

The conceptual framework links four layers: a data-integration layer consolidating fragmented data; a mapping layer representing supply-chain dependencies as a network; a modeling layer applying econometric, machine-learning, and network-risk methods; and a decision-support layer translating outputs into scenarios, indicators, and dashboards. Each layer is validated through human-in-the-loop review before informing the layer above it.

This structures Chapter Three directly: Phase 1 corresponds to the data-integration and mapping layers, Phase 2 to modeling, Phase 3 validates the decision-support layer through prototyping, and Phases 4–5 extend and update the framework at national scale.

2.9 Summary

This chapter reviewed literature on data infrastructure, supply-chain vulnerability, predictive forecasting, scenario simulation, and resilience indicators, identifying the limited availability,

among literature and tools reviewed, of an integrated, continuously updated national framework combining these methods. Chapter Three presents the methodology developed to address this gap.

CHAPTER THREE: METHODOLOGY

3.1 Research Design

This study employs a design and development research methodology, appropriate for studies whose primary contribution is a validated analytical framework rather than a single hypothesis test. The methodology is organized into five sequential but iterative phases: (1) National Trade and Industrial Intelligence Assessment, (2) Predictive Economic Framework Development, (3) Prototype Validation and Pilot Implementation, (4) National Scaling, Policy Alignment, and Dissemination, and (5) Continuous Economic Intelligence and Improvement — allowing incremental development, empirical validation before broader deployment, and continuous updating as conditions evolve.

3.2 Phase 1: National Trade and Industrial Intelligence Assessment

Phase 1 establishes the empirical foundation, assessing U.S. trade, manufacturing, supply-chain, investment, labor, energy, and macroeconomic conditions using Census trade statistics, BEA input-output and industry accounts, BLS employment and price data, USITC trade and tariff data, Federal Reserve production indices, and DOE energy data, identifying vulnerabilities, capacity constraints, and competitiveness challenges across candidate industries.

In parallel, existing federal datasets and industrial-resilience tools are catalogued along granularity, timeliness, interoperability, and lower-tier visibility, documenting precisely where Phase 2's integration work must compensate for infrastructure gaps.

The phase's core output is a supply-chain dependency map linking foreign suppliers, imported inputs, domestic manufacturers, and downstream industries, represented as a directed network weighted by transaction value and, where estimable, supplier concentration.

Phase 1 concludes with baseline Industrial Resilience and Competitiveness Indicators (supplier concentration, import dependency, capacity utilization, workforce availability, capital-investment trends) and prioritization of industries for Phase 2's predictive modeling.

3.3 Phase 2: Predictive Economic Framework Development

Phase 2 develops the integrated framework itself, combining Phase 1 data into four categories of models.

Forecasting models employ a hybrid econometric and machine-learning approach (Section 2.4): VAR and error-correction models capture structural relationships, while gradient-boosted tree and panel ML models capture nonlinear interactions; forecasts are compared and, where appropriate, combined through ensemble weighting informed by out-of-sample performance.

Supply-chain risk models quantify supplier concentration using indices such as the Herfindahl-Hirschman Index (formalized in U.S. antitrust merger-review practice; U.S. Department of Justice & Federal Trade Commission, 2023) applied to import sourcing, combined with network centrality measures from the Phase 1 dependency map.

Economic-impact models estimate how a disruption propagates through the dependency network to affect production, prices, employment, investment, and downstream industries, using input-output multiplier analysis extended with forecasting-stage elasticity estimates.

Scenario, stress-testing, and policy-simulation models integrate the above to evaluate defined scenarios — tariffs, export restrictions, geopolitical disruptions, shortages, demand shocks, and alternative interventions such as domestic-capacity investment or supplier diversification — comparing projected effects under a consistent set of assumptions.

Throughout, the framework applies data-quality controls, uncertainty analysis, model documentation, performance benchmarking against naive and existing baselines, and human-in-the-loop validation by subject-matter reviewers before outputs inform decision-support tools.

3.4 Phase 3: Prototype Validation and Pilot Implementation

Phase 3 validates the framework empirically through a pilot covering Phase 1's priority industries, developing predictive models, vulnerability-assessment tools, scenario-testing resources, and dashboards from the assembled data.

Validation proceeds two ways. Retrospective validation applies the framework to historical disruption episodes — past tariff actions, export-control episodes, shortage events — comparing retrospective forecasts against subsequently observed outcomes. Prospective stress testing applies the framework to emerging, not-yet-realized risks, generating forward-looking assessments evaluable as conditions evolve.

Performance is assessed using standard accuracy metrics (MAPE, RMSE), sensitivity analysis, and structured feedback from expert reviewers, with models refined iteratively before extension to additional industries.

Phase 3 concludes with standardized data-integration protocols, model-specification templates, and validation checklists enabling consistent, replicable application to additional industries and regions.

3.5 Phase 4: National Scaling, Policy Alignment, and Dissemination

Phase 4 extends the validated framework beyond pilot industries to additional strategically important sectors and regional economies, using Phase 3's modular components so each addition requires no redesign. This phase develops standardized national indicators for resilience, trade

exposure, capacity, concentration, and competitiveness — comparable across industries and updated recurrently (Section 2.6).

Scaling proceeds in collaboration with federal agencies, universities, professional associations, and regional economic-development stakeholders, both to validate across broader conditions and align outputs with practical stakeholder needs. Findings are disseminated through publications, technical reports, policy briefs, conferences, and workshops, with implementation guides documenting replication.

3.6 Phase 5: Continuous Economic Intelligence and Improvement

Phase 5 establishes the framework as continuously updated rather than static. New data are incorporated recurrently, and forecasts are systematically compared against realized outcomes, extending Phase 3's retrospective validation into ongoing operation. Models are recalibrated as underlying relationships evolve, avoiding the drift that otherwise degrades forecasting accuracy over time.

This phase also establishes recurring national and sector-specific stress tests and updates methodologies as new data sources and techniques emerge, maintaining relevance as conditions change.

3.7 Data Sources

The framework draws primarily on Census trade and manufacturing statistics, BEA input-output and gross-output-by-industry data, BLS employment/wage/price data, USITC tariff and trade-remedy data, Federal Reserve production indices, and DOE energy data, supplemented where accessible by industry-association reporting to improve lower-tier visibility beyond federal statistics alone. All sources, extraction dates, and transformations are documented (Section 3.3) to support reproducibility.

3.8 Modeling Approach

The modeling approach combines three traditions. Econometric time-series methods (VAR, following Sims, 1980; error-correction models; structural gravity models, following Anderson, 1979) provide interpretable, policy-facing forecasts. Machine-learning methods (gradient-boosted trees, following Friedman, 2001; random forests, following Breiman, 2001) capture nonlinear relationships, combined with econometric models through out-of-sample-informed ensemble weighting. Network and input-output methods, building on Leontief (1936) and, for cross-sector simulation, Shoven and Whalley's (1984) applied general-equilibrium tradition, represent supply-chain structure explicitly, tracing how disruption at one node propagates to others.

3.8.1 Panel Data Model Specification

The econometric core is a panel data model rather than a single cross-section or time series, appropriate for three reasons: it controls for unobserved, time-invariant industry heterogeneity; it controls for economy-wide shocks common across industries in a period; and it substantially increases usable observations relative to pure time-series or cross-sectional designs.

Two distinct specifications are presented, kept explicitly separate so the framework's full proposed architecture is never confused with the narrower model this study's empirical pilot actually estimates (Chapter Four).

Proposed National Framework Specification

The framework's full proposed specification, intended for eventual national-scale implementation once richer input-cost, tariff, and exchange-rate data are integrated, is a two-way (entity- and time-) fixed effects model:

$$Y_{it} = \alpha_i + \lambda_t + \beta_1 \text{Tariff}_{it} + \beta_2 \text{HHI}_{it} + \beta_3 \ln(\text{ForeignInputPrice})_{it} + \beta_4 \text{CapUtil}_{it} + \beta_5 \ln(\text{LaborCost})_{it} + \beta_6 \ln(\text{EnergyPrice})_{it} + \beta_7 (\text{ImportShare}_i \times \text{RER}_t) + \varepsilon_{it}$$

where i indexes industry and t indexes year; α_i and λ_t are entity and time fixed effects; Y_{it} is log domestic production; Tariff_{it} is the applicable tariff rate; HHI_{it} is import-origin concentration; $\text{ForeignInputPrice}_{it}$, LaborCost_{it} , and EnergyPrice_{it} are input-cost indices; and $\text{ImportShare}_i \times \text{RER}_t$ captures currency-movement exposure. This requires tariff, foreign-input-price, capacity, and exchange-rate data beyond this study's four integrated sources, and is not the specification estimated in Chapter Four.

Empirical Pilot Specification Estimated in This Study

Chapter Four estimates a narrower specification, using only variables constructible from the four integrated federal sources:

$$\log(Y_{it}) = \alpha_i + \lambda_t + \beta_1 \text{Employment}_{it} + \beta_2 \text{LaborProductivity}_{it} + \beta_3 \log(\text{Imports}_{it}) + \beta_4 \log(\text{Exports}_{it}) + \beta_5 \text{HHI}_{it} + \varepsilon_{it}$$

where Y_{it} is Gross Output (BEA), Employment_{it} and $\text{LaborProductivity}_{it}$ are from BLS, Imports_{it} and Exports_{it} are from USITC, and HHI_{it} is the Import-Origin Concentration Index (Section 3.3). This is the exact specification in Table 4.2; every result in Chapter Four refers to this equation, not the broader proposed specification above.

Model selection between fixed- and random-effects specifications is governed by a like-for-like Hausman (1978) test, supplemented with a Mundlak (1978) correlated random effects test (Section 4.4.1). Standard errors are entity-clustered throughout, with a wild-cluster bootstrap (Cameron, Gelbach, & Miller, 2008) given this pilot's modest 19 clusters (Section 4.4.2), following Petersen (2009).

Because manufacturing output likely exhibits genuine persistence through capacity, inventory, and order-backlog dynamics, a dynamic extension adds a lagged dependent variable:

$$\log(Y_{it}) = \rho \log(Y_{i,t-1}) + \beta' X_{it} + \alpha_i + \lambda_t + \varepsilon_{it}$$

This introduces a well-documented small-sample bias (Nickell bias) in short panels. Arellano-Bond (1991) or Blundell-Bond (1998) GMM are the standard solutions in principle, but are not applied here: with only 19 cross-sectional units, GMM risks instrument proliferation and unreliable overidentification tests. Section 4.4.5 instead reports a simpler dynamic fixed-effects specification, explicitly flagged as subject to Nickell bias, as a parsimonious check rather than a bias-corrected estimate; full GMM is future work contingent on a larger panel (Section 5.6).

3.8.2 Predictive Modeling Approach

The framework specifies a pseudo-out-of-sample, rolling-origin, one-year-ahead forecasting exercise: every predictor is lagged one year, and models are retrained at each of several rolling origins using only data available to that point, evaluated against the following year's genuinely held-out outcome. This is pseudo-out-of-sample rather than genuinely ex-ante because it uses the current, potentially revised data vintage rather than the vintage available on each historical forecast date — disclosed for methodological precision.

Origins roll forward across the panel, each trained through that year and evaluated on the following year, producing both an average figure and per-year results so readers can assess whether one year drives the average. Four approaches are compared: naive persistence; a linear model with entity fixed effects and lagged predictors; a gradient-boosted ensemble (Section 2.4); and a hybrid average.

Accuracy is evaluated using MAPE, RMSE, and MAE, benchmarked against naive persistence (Section 3.9). Reporting the naive benchmark, including where it performs competitively or better, is a required part of the methodology, not an optional check — an approach that cannot beat simple persistence should not be represented as adding forecasting value.

3.9 Validation and Evaluation Metrics

Forecast accuracy is evaluated using MAPE and RMSE against observed outcomes, benchmarked against naive forecasts and, where available, existing published forecasts. Supply-chain risk indicators are validated by comparing model-flagged high-risk inputs against documented historical disruptions. Scenario outputs are validated retrospectively against past disruption episodes (Section 3.4). All results, including shortfalls against benchmarks, are documented transparently as part of the model-governance record.

3.9.1 Human-in-the-Loop Validation Protocol

This section specifies the human-in-the-loop protocol referenced elsewhere as a design principle. It is proposed for pilot implementation, requiring the institutional partnership described in Section 3.5, and is documented here as a specification for future work, not a process already running.

Every output intended for external distribution would pass through two review roles: a methods reviewer confirming specification, diagnostics, and data-construction decisions are sound; and a domain reviewer (economists, policy analysts, or manufacturing subject-matter experts) assessing substantive plausibility beyond what the statistical model captures. Three conditions would trigger review before release: a forecast exceeding a pre-specified magnitude threshold relative to historical volatility; a regression update whose coefficients shift materially from the prior release (paralleling Sections 4.4.1–4.4.6's specification comparisons); and any result for a subsector under a known, contemporaneous disruption where stationarity assumptions are least likely to hold.

Every reviewed output would be logged with reviewer identity, date, and rationale, creating an auditable record consistent with this framework's open-source reproducibility principle (Section 5.8). This is a lighter-weight process than formal financial-sector model-risk management, scaled to what a research-pilot team could realistically operate; national-scale deployment (Phase 4) would require more formal governance.

3.10 Ethical Considerations and Limitations

The framework relies exclusively on publicly available or appropriately licensed data; any industry-level data used for validation is aggregated or anonymized. As a decision-support tool rather than an autonomous policy engine, it is designed to inform, not replace, expert judgment — the human-in-the-loop requirement in Phases 2 and 3 ensures outputs are reviewed before informing decisions.

Methodologically, the framework is subject to the general limitations of econometric and machine-learning forecasting: sensitivity to data quality, reduced accuracy around unmodeled structural breaks, and the inherent uncertainty of forward-looking models. These are addressed, though not eliminated, through the uncertainty-analysis and continuous-recalibration practices in Phases 2, 3, and 5, and discussed further in Chapter Five.

CHAPTER FOUR: EMPIRICAL RESULTS

4.1 Introduction

This chapter reports empirical results generated entirely from real, primary-source federal data, following the methodology described in Chapter Three. A panel dataset covering 19 NAICS-3 manufacturing subsectors observed annually from 2015 through 2025 (N = 209 industry-year observations) was assembled from four federal sources: BEA (Gross Output by Industry), BLS (Current Employment Statistics and Industry Productivity), and USITC (DataWeb imports, exports, and country-of-origin data used to construct an Import-Origin Concentration Index). Full source citations and construction notes are in the accompanying data workbook and Appendix A.

The forecasting exercise (Section 4.5) is described as pseudo-out-of-sample rather than genuinely ex-ante, since it uses today's data vintage rather than the vintage available on each historical forecast date; the Hausman test (Section 4.4.1) is estimated like-for-like; inference for the main regression includes a wild-cluster bootstrap given the panel's modest number of clusters (Section 4.4.2); and the concentration variable is named precisely to describe what it measures. Every result, significant or not, is reported directly.

Section 4.2 documents data-infrastructure gaps (RQ1). Section 4.3 reports descriptive statistics (RQ2). Section 4.4 reports the main panel regression and its robustness and diagnostic extensions (RQ2, RQ4). Section 4.5 reports the forecasting exercise (RQ3). Section 4.6 reports the scenario analysis (RQ5). Section 4.7 describes the decision-support interface (RQ6). Section 4.8 maps these results against the five-phase methodology from Chapter Three.

4.2 Documented Data-Infrastructure Gaps

Consistent with RQ1, assembling the panel directly surfaced the data-infrastructure limitations identified in Section 2.2, reported here as an empirical finding rather than a methodological aside.

Four gaps were documented. First, no single federal source integrates output, employment, productivity, and trade; each variable required a separate query against a separate agency system (BEA, BLS's two separate programs, USITC), with no shared access point. Second, classification granularity is inconsistent: BEA and BLS report some NAICS-3 categories only in combined form (e.g., Food and Beverage/Tobacco, NAICS 311/312), while USITC trade data is finer-grained, requiring upward aggregation and a documented approximation (Appendix A). Third, USITC reports Transportation Equipment (NAICS 336) as one combined trade figure, while BEA/BLS separate Motor Vehicles from Other Transportation Equipment, requiring a second documented approximation (employment-share splitting). Fourth, the country-of-origin detail needed for concentration analysis required a substantially larger, separate query (nearly 40,000

rows) than headline trade statistics, indicating that lower-tier visibility, while technically available, requires deliberate additional effort.

These gaps answer RQ1 directly: the limitation is fragmentation and inconsistent granularity, not an absence of underlying data. Each was resolved here through disclosed construction decisions, but each represents a standing barrier any national-scale implementation would need to resolve systematically rather than ad hoc.

4.3 Panel Dataset and Descriptive Statistics

Table 4.1 reports descriptive statistics across all 209 observations, addressing RQ2. The panel spans subsector sizes from \$15.7B (Apparel) to \$1,268.3B (Food and Beverage/Tobacco) in gross output, and import-origin concentration from HHI 646 to 3,606. A full year-by-subsector snapshot is in Appendix A.

Table 4.1. Descriptive Statistics, Full Panel (N = 209; 19 NAICS-3 Manufacturing Subsectors × 11 Years, 2015–2025)

Statistic	Gross Output (\$B)	Employment (000s)	Exports (\$B)	Imports (\$B)	Labor Productivity (Index)	Import-Origin HHI (0–10,000)
Mean	330.1	658.1	76.0	124.6	97.9	1,558.4
Std. Deviation	287.5	462.3	72.3	115.0	6.0	672.9
Minimum	15.7	79.0	4.4	4.9	78.6	645.9
Median	231.4	427.4	59.4	94.9	99.3	1,481.8
Maximum	1,268.3	2,098.7	302.7	709.4	115.0	3,605.9

This variable is termed the Import-Origin Concentration Index rather than "Supplier Concentration" to describe precisely what it measures: concentration of import value among countries of origin (Herfindahl-Hirschman methodology, Section 3.2), computed from USITC data. It is a macro-level, country-of-origin proxy, not a firm-level supplier measure, and does not specifically isolate critical inputs — it reflects all trade value classified within a subsector's USITC codes. In 2025, Food and Beverage/Tobacco showed the lowest concentration (HHI 837); Wood Products and Printing showed the highest (above 1,700).

A measurement caveat: Gross Output, Imports, and Exports are in nominal (current) dollars, not deflated. Year fixed effects absorb economy-wide inflation but not subsector-specific price movements. This is disclosed as a limitation rather than corrected here; a real (deflated) panel is a priority for future work (Section 5.6).

4.4 Panel Regression: Explaining Manufacturing Output

The two-way fixed-effects specification (Section 3.8.1) was estimated with $\log(\text{Gross Output})$ as the dependent variable and Employment, Labor Productivity, $\log(\text{Imports})$, $\log(\text{Exports})$, and standardized Import-Origin Concentration as regressors. Standard errors are entity-clustered; because the panel has only 19 clusters, a wild-cluster bootstrap (Section 4.4.2) is the more credible inferential statement than the asymptotic p-values below.

Table 4.2. Two-Way Fixed Effects Panel Regression — Dependent Variable: $\log(\text{Gross Output})$; $N = 209$; entity-clustered standard errors (asymptotic; see Section 4.4.2 for wild-cluster-bootstrap inference)

Variable	Coefficient	Std. Error	t-stat	Asymptotic p-value
Employment (thousands)	0.0002	0.0002	0.85	0.397 (n.s.)
Labor Productivity (Index, 2017=100)	0.0005	0.0018	0.30	0.765 (n.s.)
$\log(\text{Imports})$	0.1998	0.1023	1.95	0.052
$\log(\text{Exports})$	0.2747	0.1349	2.04	0.043
Import-Origin Concentration (HHI, standardized)	-0.0240	0.0220	-1.09	0.275 (n.s.)

Model fit: R^2 (within) = 0.619; $F(5, 175) = 17.96$, $p < 0.001$. Under asymptotic clustering, exports are significant and imports marginally so; under the wild-cluster bootstrap (Section 4.4.2), neither reaches significance (imports $p = 0.142$; exports $p = 0.078$). The honest conclusion is more conservative: trade activity is positively associated with output, but this panel does not provide statistically decisive evidence of that association under small-sample-appropriate inference. These coefficients reflect co-movement, not a causal trade elasticity. Employment and Labor Productivity are not significant; Import-Origin Concentration is negative but not significant under either inference approach ($p = 0.275$ asymptotic, $p = 0.263$ bootstrap).

4.4.1 Robustness Checks: Alternative Model Specifications and a Like-for-Like Hausman Test

Four alternative specifications were estimated on the identical panel: Pooled OLS, Random Effects, Two-Way FE excluding concentration, and Entity-FE-only.

Table 4.3. Alternative Model Specifications, Dependent Variable: $\log(\text{Gross Output})$; $N = 209$

Variable	Pooled OLS	Random Effects	Two-Way FE (no HHI)	Two-Way FE (Main Model)	Entity FE only
Employment	0.0007**	0.0005**	0.0001 (n.s.)	0.0002 (n.s.)	0.0003 (n.s.)
Labor Productivity	-0.0193**	-0.0013 (n.s.)	0.0007 (n.s.)	0.0005 (n.s.)	-0.0010 (n.s.)
$\log(\text{Imports})$	-0.5432**	0.1983*	0.1934*	0.1998*	0.2824**

Variable	Pooled OLS	Random Effects	Two-Way FE (no HHI)	Two-Way FE (Main Model)	Entity FE only
log(Exports)	0.8430**	0.4078**	0.2526*	0.2747**	0.3611**
Import-Origin Concentration (HHI, std.)	-0.0721*	-0.0764**	—	-0.0240 (n.s.)	-0.0679**
R ² (overall / within)	0.861	0.693 / 0.704	0.329 / 0.553	0.339 / 0.619	0.710

A Hausman test comparing Random Effects against Fixed Effects is only valid on a like-for-like basis — both models must share the same year-effect structure. Table 4.4 reports a corrected, like-for-like Hausman test (Random Effects re-estimated with year dummies) alongside a Mundlak (correlated random effects) test.

Table 4.4. Model-Selection Tests: Fixed Effects vs. Random Effects

Test	Statistic	p-value	Conclusion
Hausman test (like-for-like, year effects in both models)	$\chi^2(5) = 5.36$	0.374	Fails to reject Random Effects
Mundlak test (joint significance of entity-mean regressors)	$\chi^2(5) = 48.97$	< 0.0001	Rejects Random Effects; supports Fixed Effects

These tests point in different directions and both are reported. The Hausman test alone does not decisively favor Fixed Effects. The Mundlak test, which directly tests whether entity-level averages are correlated with the unobserved individual effect, does support Fixed Effects, and is treated as the more directly relevant test here. Fixed Effects remains the preferred specification as a reasoned judgment from mixed evidence, not a test result that settles the question beyond doubt.

The Import-Origin Concentration coefficient is negative and significant in three of the four alternative specifications, losing significance only in the preferred two-way fixed-effects model. It would be tempting to treat this as evidence the relationship is robust — but once two-way FE is identified as the preferred specification, significance in less-controlled alternatives does not strengthen the main result; if anything, losing significance once common year shocks are controlled for suggests the apparent relationship in weaker specifications may partly reflect shared trends. The honest conclusion: Import-Origin Concentration has a consistently negative coefficient across specifications, but the preferred specification does not provide statistically conclusive evidence of a within-industry effect.

4.4.2 Additional Diagnostics

Additional diagnostic tests were run on the real panel and are reported regardless of outcome.

Table 4.5. Additional Diagnostic Tests

Diagnostic	Result	Interpretation
Wild-cluster bootstrap (999 reps, cluster = entity) — log(Imports)	statistic = 1.91, $p = 0.142$	Not significant at 5%; weaker than the asymptotic result
Wild-cluster bootstrap — log(Exports)	statistic = 1.99, $p = 0.078$	Not significant at 5%, marginal at 10%
Wild-cluster bootstrap — Import-Origin Concentration	statistic = -1.07, $p = 0.263$	Not significant, consistent with asymptotic
Variance Inflation Factors (all regressors)	range 1.08–4.41	Well below conventional concern thresholds
Pesaran CD test (cross-sectional dependence)	CD = -1.38, $p = 0.169$	Not significant; supports entity-only clustering
Wooldridge-style serial correlation test (H_0 : coefficient = -0.5)	$t = 8.65$, $p < 0.0001$	Rejects the null; evidence of persistence a static model misses

Three findings matter. First, the wild-cluster bootstrap — more credible than large-cluster asymptotics for a 19-entity panel — finds neither trade coefficient significant at 5%, revising the paper's conclusions accordingly: trade activity is positively associated with output, but not decisively so under appropriate inference. Second, the absence of cross-sectional dependence supports the entity-only clustering used throughout. Third, the serial correlation test rejects its null, indicating persistence a static model misses — addressed directly with a dynamic robustness check in Section 4.4.5.

4.4.3 Transportation-Equipment Allocation Sensitivity Check

Section 4.2 discloses an approximation: USITC's combined NAICS 336 trade figure is split between Motor Vehicles and Other Transportation Equipment by employment share. Table 4.5a re-estimates the main specification three ways — current allocation, omitting both affected subsectors, and collapsing them into one category matching USITC's native reporting.

Table 4.5a. Main Specification Under Three Transportation-Equipment (NAICS 336) Allocation Approaches

Variable	Current allocation (main, N=209)	Omit both transport subsectors (N=187)	Collapsed to one category (N=198)
log(Imports)	0.1998 ($p=0.052$)	0.2064 ($p=0.071$)	0.2008 ($p=0.059$)
log(Exports)	0.2747 ($p=0.043$)	0.2809 ($p=0.072$)	0.2720 ($p=0.058$)
Import-Origin Concentration (HHI)	-0.0240 ($p=0.275$)	-0.0286 ($p=0.292$)	-0.0257 ($p=0.282$)

All three coefficients remain close in sign and magnitude; the transportation-equipment allocation approximation does not appear to drive the paper's results.

4.4.4 Trade × Concentration Interaction

The baseline model cannot distinguish whether concentrated subsectors are more vulnerable to a given trade shock than diversified ones. Table 4.6 tests this directly with an Imports × Concentration interaction term.

Table 4.6. Two-Way Fixed Effects Model with Trade × Concentration Interaction

Variable	Coefficient	Std. Error	t-stat	p-value
Employment	0.0003	0.0002	1.08	0.280 (n.s.)
Labor Productivity	0.0011	0.0016	0.64	0.524 (n.s.)
log(Imports)	0.2018	0.1033	1.95	0.052
log(Exports)	0.2933	0.1386	2.12	0.036
Import-Origin Concentration (HHI, std.)	-0.3207	0.6642	-0.48	0.630 (n.s.)
log(Imports) × Concentration (HHI, std.)	0.0119	0.0265	0.45	0.653 (n.s.)

The interaction term is not significant ($p = 0.653$): this panel does not find evidence that highly concentrated subsectors are more sensitive to import fluctuations than diversified ones. The absence of significance may reflect limited statistical power, model specification, measurement limitations, or a genuine absence of the effect — this panel cannot distinguish among these explanations, and re-testing with a larger panel is a priority (Section 5.6).

4.4.5 Dynamic Panel Robustness Check

Section 4.4.2's serial-correlation diagnostic found persistence a static model misses. Table 4.5b reports a parsimonious dynamic check: the specification extended with a one-period lagged dependent variable. This is explicitly not an Arellano-Bond or Blundell-Bond GMM estimate — with only 19 entities, GMM risks instrument proliferation — and is subject to a well-documented small-sample bias (Nickell bias) in short panels, reported as a robustness check, not a bias-corrected structural estimate.

Table 4.5b. Dynamic Panel Specification with Lagged Dependent Variable ($N = 190$; one year lost to lagging); asymptotic entity-clustered inference

Variable	Coefficient	Std. Error	t-stat	Asymptotic p-value	Wild-bootstrap p-value
log(Gross Output), lagged one year	0.4149	0.1927	2.15	0.033	0.063

Variable	Coefficient	Std. Error	t-stat	Asymptotic p-value	Wild-bootstrap p-value
Employment	0.0001	0.0002	0.83	0.408 (n.s.)	0.405 (n.s.)
Labor Productivity	0.0009	0.0016	0.58	0.561 (n.s.)	0.613 (n.s.)
log(Imports)	0.2113	0.1176	1.80	0.074	0.233 (n.s.)
log(Exports)	0.1270	0.1080	1.18	0.241 (n.s.)	0.256 (n.s.)
Import-Origin Concentration (HHI, std.)	-0.0257	0.0196	-1.32	0.190 (n.s.)	0.260 (n.s.)

The lagged output term's asymptotic coefficient ($p = 0.41$) is consistent with substantial persistence, but under the small-sample-appropriate wild-cluster bootstrap it is only marginally significant ($p = 0.063$ vs. 0.033 asymptotically) — suggestive but not decisive evidence of persistence.

Imports and concentration move only modestly from the static specification (Table 4.2), but the export coefficient falls from 0.2747 to 0.1270 — roughly half its static value — losing its marginal significance. This suggests part of the static model's export-output association reflects output's own persistence rather than a stable contemporaneous relationship. Both specifications are reported so the reader can see how sensitive the finding is to this choice.

4.4.6 First-Differenced Robustness Check

Output, Imports, and Exports are nominal, not deflated (Section 4.3); since all three trend upward, part of the main specification's associations could in principle reflect shared trends. As an additional check, the specification was re-estimated in first differences. To be precise: differencing alone removes each entity's fixed effect, but an entity-specific linear trend survives as an entity-specific constant — removing that requires including entity dummies in the differenced regression itself, which this specification does. A wild-cluster bootstrap is again reported alongside the asymptotic result.

Table 4.5c. First-Differenced Specification with Entity and Year Effects (N = 190; one year lost to differencing)

Variable	Coefficient (main FE, Table 4.2)	Coefficient (first-differenced)	Asymptotic p-value	Wild-bootstrap p-value
Employment	0.0002	-0.0001	0.896 (n.s.)	0.931 (n.s.)
Labor Productivity	0.0005	0.0015	0.193 (n.s.)	0.161 (n.s.)
log(Imports)	0.1998	0.2124	0.294 (n.s.)	0.556 (n.s.)
log(Exports)	0.2747	0.2499	0.019	0.001
Import-Origin Concentration (HHI, std.)	-0.0240	-0.0447	0.225 (n.s.)	0.503 (n.s.)

With entity-specific trends properly removed, imports lose significance entirely under both inference approaches. Exports remain significant, close in magnitude to the main result (0.2747 vs. 0.2499), and notably under the wild-cluster bootstrap ($p = 0.001$) — the only trade coefficient in this chapter that remains significant under small-sample-appropriate inference once trends are controlled for. This does not resolve the nominal-versus-real measurement limitation (Section 4.3), but identifies the export-output association specifically, not the trade-output association generally, as the most robust single finding in this chapter.

4.4.7 Employment-Impact Model

Sections 4.4–4.4.6 focus on output. This section extends the framework to employment specifically, controlling for output and productivity — directly addressing the goal of estimating disruption effects on production and employment, not output alone.

Table 4.5d. Employment-Impact Model — Dependent Variable: $\log(\text{Employment})$; $N = 209$; entity-clustered standard errors

Variable	Coefficient	Std. Error	t-stat	Asymptotic p-value	Wild-bootstrap p-value
$\log(\text{Gross Output})$	0.1322	0.1051	1.26	0.210 (n.s.)	0.281 (n.s.)
Labor Productivity	−0.0041	0.0025	−1.61	0.109 (n.s.)	0.213 (n.s.)
$\log(\text{Imports})$	0.1467	0.0832	1.76	0.080 (n.s.)	0.131 (n.s.)
$\log(\text{Exports})$	−0.0287	0.0983	−0.29	0.770 (n.s.)	0.820 (n.s.)
Import-Origin Concentration (HHI, std.)	0.0346	0.0217	1.60	0.112 (n.s.)	0.204 (n.s.)

None of the five regressors reach significance under either inference approach — a genuine null result, reported directly. One pattern worth noting despite the lack of significance: the concentration coefficient is positive here, the opposite sign from its output association, consistent with a plausible (but statistically unconfirmed) story that more concentrated subsectors may be somewhat more labor-intensive per unit of output. Establishing genuine employment-impact estimates is a priority for a larger panel (Section 5.6).

4.5 Predictive Model: Pseudo-Out-of-Sample Forecast Evaluation

A forecasting exercise that uses a period's actual realized predictor values to predict that period's output is conditional prediction (nowcasting), not ex-ante forecasting, since those values would not genuinely have been available at forecast time. This exercise avoids that by lagging every predictor one year, so only information available at $t-1$ predicts t .

The model is retrained at each of four origin years (2021–2024) using only data available up to that point, forecasting exactly one year ahead each time — a rolling-origin design (see Section 3.8.2 for why "pseudo-out-of-sample" rather than "genuinely ex-ante" is the precise term, given current rather than historical data vintages are used). Concentration is standardized using only each origin's training window. Four approaches are compared: naive persistence; linear (entity FE + trend, lagged); gradient boosting; and a hybrid average.

Table 4.7. Pseudo-Out-of-Sample Rolling-Origin Forecast Accuracy (MAPE), by Individual Forecast Year and Average

Model	2022	2023	2024	2025	Average MAPE	Avg. RMSE (\$B)	Avg. MAE (\$B)
Naive persistence	10.19%	5.06%	2.53%	1.97%	4.94%	37.43	20.68
Linear (entity FE + trend, lagged)	13.44%	6.34%	6.18%	6.22%	8.04%	49.85	30.18
Gradient Boosting (lagged)	15.70%	14.13%	12.80%	4.99%	11.90%	118.91	58.05
Hybrid (Linear + GBR average)	12.18%	9.86%	7.61%	4.40%	8.51%	82.66	40.80

The naive persistence benchmark achieves the lowest average error of all four approaches by a wide margin (4.94% vs. 8.04% or worse for every model-based approach), reported directly. With genuinely available lagged predictors at this panel's size, neither the econometric nor the machine-learning approach adds forecasting value over simply carrying forward each subsector's most recent output.

This carries a direct implication for the proposed framework: one-year-ahead forecasting with this panel's covariates is genuinely difficult, and a deployed version should be benchmarked against, and should expect to sometimes underperform, a simple persistence forecast — precisely the evaluation discipline specified in Section 3.9.

4.6 Model-Implied Scenario Analysis: Illustrative, Non-Causal Trade Disruption Effects

This section uses Table 4.2's fitted coefficients to compute model-implied scenarios: the predicted percentage output change from a hypothetical import or concentration shift, holding other factors constant. This is described as an Exploratory Model-Implied Scenario Tool, not a causal policy simulator, and every result should be read as an illustrative association, not a causal

forecast. Genuine causal estimation would require exogenous variation — natural experiments, shift-share instruments, event-study or difference-in-differences designs, or a structural CGE model — none of which this exercise provides.

For a subsector with import elasticity 0.1998 (illustrative despite not being statistically decisive under bootstrap inference), a hypothetical X percent import reduction implies a model-implied output change of $(1+X)^{0.1998} - 1$.

Table 4.8. Illustrative Model-Implied Output Association with a Hypothetical Import Disruption, by Subsector and Shock Magnitude (2025 Baseline) — Exploratory and Non-Causal

Subsector	2025 Baseline Output (\$B)	–10% Imports	–20% Imports	–30% Imports
Computer and electronic products	425.6	\$416.7B (–2.08%)	\$407.0B (–4.36%)	\$396.3B (–6.88%)
Chemical products	1,016.7	\$995.5B (–2.08%)	\$972.4B (–4.36%)	\$946.8B (–6.88%)
Food and beverage and tobacco products	1,268.3	\$1,241.9B (–2.08%)	\$1,213.0B (–4.36%)	\$1,181.1B (–6.88%)
Primary metals	303.4	\$297.1B (–2.08%)	\$290.2B (–4.36%)	\$282.5B (–6.88%)

This arithmetic implies an identical percentage association for a given shock across every subsector, differing only in dollar terms; Section 4.4.4's interaction test found no significant evidence that concentrated subsectors are more sensitive, so this uniform-elasticity property is consistent with what the data show, within this panel's statistical power. A parallel example: a one-standard-deviation concentration improvement (–673 HHI points) implies a model-implied output change of approximately 2.37%; for a median-sized subsector (~\$231B), roughly \$5.5 billion — presented here as an illustration of the calculation, not a statistically supported benefit estimate, since the underlying coefficient is not significant in the preferred specification (Section 4.4).

4.6.1 Comparative Illustration Across Alternative Interventions

The framework's design calls for comparing alternative interventions, not evaluating a single shock in isolation. Table 4.8a applies Table 4.2's coefficients to three named interventions for Computer and Electronic Products (highest import exposure): a tariff-style import reduction, a supplier-diversification action, and a combined approach.

Table 4.8a. Illustrative Comparison of Alternative Interventions — Computer and Electronic Products, 2025 Baseline (\$425.6B); Exploratory and Non-Causal

Intervention	Assumption	Model-Implied Output Change	Resulting Output
A: Tariff-style import reduction	–20% imports	–4.36%	\$407.0B
B: Supplier	–1 SD import-origin	+2.43%	\$435.9B

Intervention	Assumption	Model-Implied Output Change	Resulting Output
diversification	concentration		
C: Combined approach (A + B)	Both simultaneously	-2.04%	\$416.9B

This illustrates a trade-off the fitted model implies — import reduction and diversification pull output in opposite directions — but two caveats apply. The concentration coefficient underlying Scenario B is not significant in the preferred specification, so its contribution is illustrative, not a supported claim. And this arithmetic treats the two interventions as independent and additive, when reducing concentration would plausibly also change import volumes. A genuine policy-comparison capability resolving this with statistical and causal confidence is future work (Section 5.6).

4.7 Decision-Support Interface

Addressing RQ6, a prototype decision-support dashboard (the Economic Intelligence Copilot) draws on this real panel, presenting summary indicators, output trends, the results in this chapter, and the Exploratory Model-Implied Scenario Tool, labeled consistently with this chapter's non-causal framing throughout. The interface has not been formally evaluated against alternative formats, so RQ6 is only partially answered; systematic evaluation with real policymaker or manufacturer users is a priority for future research (Chapter Five).

4.8 Correspondence with the Proposed Five-Phase Methodology

This chapter's empirical work constitutes real evidence touching the first three phases of Chapter Three's methodology, but does not validate every technique in that broader architecture. Structural gravity models, full VAR systems, network-based dependency mapping, and CGE modeling are not implemented here; the methods actually estimated are a two-way fixed-effects regression and its robustness variants, a rolling-origin forecasting exercise, a computed concentration index, and arithmetic scenario calculations. Table 4.9 states plainly what has and has not been demonstrated.

Table 4.9. Correspondence Between Empirical Work Completed and the Proposed Five-Phase Methodology

Phase	Demonstrated with real data	Not yet demonstrated
Phase 1: National Trade and Industrial Intelligence Assessment	Integration of real federal data across BEA, BLS, and USITC; documented data-infrastructure gaps (Section 4.2); baseline resilience and concentration indicators (Section 4.3)	A formal dependency network linking specific foreign suppliers to specific domestic manufacturers and downstream industries
Phase 2: Predictive Economic Framework Development	A two-way fixed-effects econometric model with honestly	The gravity-model, VAR, and CGE components described in Section

Phase	Demonstrated with real data	Not yet demonstrated
	reported inference (Sections 4.4–4.4.7); a pseudo-out-of-sample, rolling-origin forecasting exercise (Section 4.5); a real, computed import-origin concentration index; an employment-impact model, honestly reporting a null result (Section 4.4.7); a comparative illustration across named alternative interventions (Section 4.6.1); a documented human-in-the-loop validation protocol specified for pilot implementation (Section 3.9.1)	3.8 as part of the framework's intended architecture; economic-impact models estimating disruption effects on prices specifically (only employment was estimated); the human-in-the-loop protocol operated in practice rather than only specified; a policy-comparison capability that resolves trade-offs with statistical and causal confidence rather than illustrating them arithmetically
Phase 3: Prototype Validation and Pilot Implementation	A decision-support dashboard interface drawing on the real panel (Section 4.7); an out-of-sample forecast evaluation against a naive benchmark (Section 4.5); a battery of regression diagnostics (Section 4.4.2)	Retrospective validation of model predictions against specific historical disruption episodes; formal evaluation of decision-support interface formats (Section 4.7); a standardized, documented procedure for replicating the framework across additional industries
Phases 4–5: National Scaling and Continuous Improvement	Not applicable at this stage	These phases are institutional and programmatic in nature and are outside the scope of a single empirical analysis

This correspondence is an honest progress statement, not a claim of completion. The results demonstrate the framework's core data-integration and modeling concepts are empirically workable, including honest demonstrations that a pseudo-out-of-sample forecast is difficult to improve on a naive benchmark, that concentration-output evidence is directionally consistent but not statistically decisive, and that differential vulnerability by concentration level is not currently supported. The gravity-model, network-mapping, VAR, and CGE components remain future work.

4.9 Summary of Findings

This chapter reported results from a real panel of 19 U.S. manufacturing subsectors, 2015–2025, organized around the six research questions from Chapter One. Data-infrastructure gaps were documented directly (RQ1). A two-way fixed-effects regression found trade activity positively associated with output under asymptotic inference, though not significant under the more appropriate wild-cluster bootstrap; import-origin concentration was consistently negative but not significant in the preferred specification (RQ2, RQ4). An employment-impact model found no

significant relationship at all — a genuine null result (RQ2, RQ4). A pseudo-out-of-sample forecasting exercise found a naive benchmark difficult to outperform (RQ3). A trade × concentration interaction was not significant; a comparative illustration across three named interventions demonstrated the kind of policy comparison the framework is designed to eventually support (RQ5). A prototype dashboard was developed but not formally evaluated, and a human-in-the-loop protocol was specified but not yet operated (RQ6). Findings — including non-significant ones that qualify the paper's headline claims — are disclosed directly throughout. Chapter Five discusses implications, contributions, limitations, and future research.

CHAPTER FIVE: DISCUSSION, CONCLUSIONS, AND RECOMMENDATIONS

5.1 Discussion of Findings

The empirical results in Chapter Four support this study's central premise while disciplining it with honestly reported evidence, including findings that qualify rather than strengthen several headline claims. The panel regression (Section 4.4) finds trade activity positively associated with output under asymptotic inference, but the wild-cluster bootstrap (Section 4.4.2) — the more appropriate approach for this panel's 19 entity clusters — does not find that association significant. Import-origin concentration is consistently negative across specifications, significant in three of four alternatives (Table 4.3), but not in the preferred two-way fixed-effects or interaction specification (Section 4.4.4); as Section 4.4.1 explains, significance in less-controlled alternatives does not rescue the preferred specification's null result. The concentration-risk hypothesis is neither confirmed nor refuted; the study identifies precisely what a larger panel would need to establish it more decisively.

The pseudo-out-of-sample forecasting exercise (Section 4.5) provides a second discipline: a naive persistence benchmark is difficult to outperform. This is not a negative result — it is the honest benchmark comparison Section 3.9 requires, demonstrating the framework's evaluation standards are applied rigorously, not merely stated.

5.2 Contributions to the Field

This study contributes in four ways. First, a documented, replicable methodology for integrating fragmented federal and industry data (Section 2.2). Second, an architecture integrating network-based dependency mapping with econometric forecasting (Section 2.3), though the formal dependency-network component remains ongoing development, not an implemented and validated method (Table 4.9). Third, an honestly evaluated hybrid econometric/machine-learning forecasting approach, including a pseudo-out-of-sample evaluation finding it does not currently outperform a naive benchmark. Fourth, a prototype comparative scenario methodology and national indicators, explicitly illustrative and non-causal, addressing the tool-usability gap (Section 2.7).

Collectively, these demonstrate an integrated applied framework combining established methods — panel econometrics, machine-learning forecasting, concentration measurement, decision-support presentation — commonly implemented separately in this domain, extensible through the Phase 4/5 methodology well beyond this study's pilot scope.

5.3 Policy Implications and National Significance

This section addresses why further pilot development is warranted, what economic benefits can reasonably be expected, and how the framework bears on competitiveness and resilience — grounded in Chapter Four's real results, not design intentions alone.

5.3.1 Policy Relevance and Case for Further Pilot Implementation

Three findings support further pilot development. First, the framework is buildable from data policymakers already have: every variable came from existing, already-funded federal systems, requiring an integration capability rather than new data collection. Second, the diagnostic work (Sections 4.4.1–4.4.2) shows the evaluation apparatus itself is rigorous — a like-for-like Hausman test, a Mundlak test, a wild-cluster bootstrap, and multiple specifications reported side by side, including results that do not support the framework's central hypotheses — a more reliable property than any single estimate. Third, the forecasting exercise (Section 4.5), despite finding naive persistence hard to beat, demonstrates the evaluation discipline (Section 3.9) would catch an overconfident forecasting claim before it reached a policymaker — exactly the safeguard a national decision-support tool should have.

Beyond these findings: among the federal systems examined in this study (BEA's, BLS's, USITC's own interfaces), none integrates output, employment, productivity, and trade-concentration data into a single recurring panel — agencies produce these separately, requiring manual reconciliation (Section 4.2). This reflects the systems examined, not an exhaustive survey. Institutionalizing that integration, even at this pilot's 19-subsector scale, would save recurring analytical effort. This need is not hypothetical: the current Trade Policy Agenda identifies securing supply chains for critical minerals and strategic sectors as an explicit national priority (Office of the United States Trade Representative, 2026), underscoring this framework's direct responsiveness to a stated, current policy objective.

5.3.2 Potential Economic Relevance and Analytical Efficiency

The kind of estimate this framework could eventually produce is illustrated in Section 4.6's scenario tool; a worked dollar-value example appears there rather than here, since the underlying concentration coefficient is not significant in the preferred specification and should not be presented as an expected benefit. A genuine, decision-grade estimate is future work (Section 5.6), contingent on a larger panel.

What can be stated with more confidence is analytical efficiency: the data-integration work (Section 4.2), once built, is reusable. Assembling 209 real observations was a substantial one-time cost; a recurring, institutionalized version would not repeat it for every new policy question, and each additional year or subsector (Phase 4, Section 3.5) would be cheaper to incorporate than the

first — a front-loaded-cost, low-marginal-cost structure typical of successful federal data infrastructure investments.

5.3.3 Impact on Manufacturing Competitiveness and Supply-Chain Resilience

The panel itself (Appendix A) already demonstrates diagnostic value before any policy action. The 2025 cross-section identifies Wood Products (HHI 1,962) and Printing (HHI 1,736) as most import-concentrated, and Food and Beverage/Tobacco (HHI 837) as most diversified — a comparative exposure picture available directly, independent of the concentration-output coefficient's significance, positioning the framework as a starting point for identifying subsectors warranting closer monitoring.

For competitiveness: imports and exports both have positive coefficients (Table 4.2), though neither is significant under the preferred bootstrap inference (Section 4.4.2) — suggestively, not statistically, indicating subsectors more integrated into global trade may be more productive. This is a hypothesis for further testing, not an established finding; competitiveness policy should weigh import-side integration alongside export promotion rather than treating imports as uniformly adverse.

For resilience: the trade $\times$ concentration interaction test (Section 4.4.4) found no significant evidence that concentrated subsectors are more shock-sensitive — a genuine null result, reported honestly rather than omitted. Policymakers using any current-generation trade-risk tool should treat uniform-elasticity assumptions with caution; this study can state that caveat credibly because it tested the interaction directly rather than only asserting the limitation.

5.4 Recommendations for Implementation

Four recommendations follow from this study's findings and design experience. First, prioritize data-integration work before predictive modeling, since forecasting and scenario-simulation accuracy depend on the underlying data's quality (Section 3.2). Second, treat lower-tier supply-chain visibility as a standing priority, not a one-time exercise — the current country-level concentration index cannot capture firm-level or multi-tier dependencies. Third, maintain human-in-the-loop validation as a permanent governance feature, not a transitional step, given forecasting's inherent uncertainty (Section 3.10). Fourth, implement industry-by-industry using Phase 3's modular components rather than simultaneous national deployment, allowing validation at each stage.

5.5 Limitations of the Study

This study has several limitations. Chapter Four's estimates are genuine empirical results from real federal data, not simulated figures — but they are pilot-scale results from a modest 19-

subsector, 11-year panel, not a comprehensive national deployment, causal estimates, or fully generalizable beyond this panel's scope. National-scale deployment would require a substantially larger panel and, for causal claims, a design isolating exogenous variation.

The predictive models share the general limitations of econometric and machine-learning forecasting (Section 3.10): reduced accuracy around unmodeled structural breaks and sensitivity to data quality. Lower-tier visibility remains unaddressed by the current concentration measure. The 19-subsector pilot scope, while spanning a range of dependency structures, cannot capture the full diversity of U.S. manufacturing; Phase 4 scaling (Section 3.5) would be required to establish broader performance.

5.6 Recommendations for Future Research

Future research should pursue fuller empirical implementation using a larger cross-section, with systematic retrospective validation against more historical disruption episodes. The non-significant but correctly signed concentration result (Section 4.4), the non-significant interaction (Section 4.4.4), and naive persistence's difficulty to beat (Section 4.5) all point to the same need: a larger panel would sharpen statistical power for hypotheses that are directionally consistent with theory but not yet decisive at this scale. Further research should pursue formal dependency-network mapping (Section 2.3; Table 4.9), extend the fixed-effects specification to genuine price- and employment-impact models, and examine dynamic panel GMM methods (Section 3.8.1) to address the serial correlation documented in Section 4.4.2. A real (deflated) version of this panel, using subsector-specific price indices, is a priority given the current panel's nominal dollar values (Section 4.3). Finally, research on governance arrangements to sustain Phase 5's continuous-update process — data-sharing agreements among federal agencies, universities, and industry — would support long-term national scaling.

5.7 Conclusion

This study set out to address a documented gap in U.S. trade and industrial-policy analysis: the limited availability, among existing tools and literature, of an integrated, predictive, supply-chain-aware framework translating fragmented federal data into forward-looking insight. Building on Chapter Two's literature, Chapter Three developed a five-phase methodology integrating data assembly, dependency mapping, hybrid forecasting, and scenario simulation, demonstrated through Chapter Four's pilot-scale empirical validation on real federal data.

The resulting Predictive Trade and Industrial Intelligence Framework offers a replicable methodology for strengthening the analytical foundation of U.S. trade policy, manufacturing resilience, and industrial competitiveness — of direct and continuing relevance to U.S. national economic interest. While full national implementation remains a substantial undertaking beyond

this study's scope, the framework, methodology, and validation approach documented here are intended as a rigorous, replicable starting point for that broader effort.

5.8 Data and Code Availability

The complete panel dataset, the full analysis pipeline (data-construction, panel regression, robustness, diagnostic, and forecasting scripts), the documented methodology, and the decision-support dashboard prototype (the Economic Intelligence Copilot) are released as an open-source repository under the MIT License at <https://github.com/aolusoji/economic-intelligence-copilot>. The repository allows independent reproduction of every table in Chapter Four directly from the included dataset, with a documented column schema so the pipeline can be pointed at a different dataset, industry set, or time period. The live decision-support interface, including the scenario tool (Section 4.6), is deployed from the same repository. Researchers extending this work are encouraged to fork the repository directly rather than requesting materials from the author.

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APPENDIX A: REAL PANEL DATASET

A.1 Dataset Description and Summary Statistics

This appendix presents descriptive statistics and a full-year snapshot of the real panel dataset underlying Sections 4.2–4.6. The complete dataset (209 industry-year observations, 2015–2025) is in the accompanying workbook, *Manufacturing_Panel_Dataset.xlsx*, with a full data-sources and construction-methodology sheet.

Table A.1 reports summary statistics across all 209 observations.

Table A.1. Descriptive Statistics, Full Panel (N = 209; 19 NAICS-3 Manufacturing Subsectors × 11 Years, 2015–2025)

Statistic	Gross Output (\$B)	Employment (000s)	Exports (\$B)	Imports (\$B)	Labor Productivity (Index)	HHI (0–10,000)
Mean	330.1	658.1	76.0	124.6	97.9	1,558.4
Std. Deviation	287.5	462.3	72.3	115.0	6.0	672.9
Minimum	15.7	79.0	4.4	4.9	78.6	645.9
25th Percentile	124.3	368.6	11.8	36.9	93.8	987.7
Median	231.4	427.4	59.4	94.9	99.3	1,481.8
75th Percentile	412.6	944.2	118.5	168.5	101.2	1,907.6
Maximum	1,268.3	2,098.7	302.7	709.4	115.0	3,605.9

A.2 Full-Panel Snapshot: 2025

Table A.2 presents the panel's most recent year (2025) for all 19 subsectors, illustrating the cross-sectional variation the panel regression exploits. Full annual detail for 2015–2025 is in the accompanying workbook.

Table A.2. Manufacturing Subsector Panel Data, 2025

Subsector	Gross Output (\$B)	Employment (000s)	Exports (\$B)	Imports (\$B)	Labor Productivity (Index)	HHI
Apparel and leather and allied products	22.6	79.0	9.38	121.62	115.0	1,065
Chemical products	1,016.7	900.5	293.20	411.28	90.0	971
Computer and electronic products	425.6	997.7	302.68	709.35	114.5	1,299

Subsector	Gross Output (\$B)	Employment (000s)	Exports (\$B)	Imports (\$B)	Labor Productivity (Index)	HHI
Electrical equipment, appliances, and components	199.3	429.0	76.01	192.34	93.4	1,243
Fabricated metal products	464.7	1,436.2	60.74	88.11	89.2	946
Food and beverage and tobacco products	1,268.3	2,098.7	92.77	149.06	78.6	837
Furniture and related products	85.3	337.9	4.81	39.90	89.7	1,643
Machinery	437.3	1,089.1	160.21	233.46	93.7	762
Miscellaneous manufacturing	181.9	612.7	84.06	132.91	82.8	955
Motor vehicles, bodies and trailers, and parts	750.7	967.4	175.75	246.72	100.0	1,794
Nonmetallic mineral products	174.3	417.0	11.40	27.47	88.2	892
Other transportation equipment	409.2	776.1	141.00	197.94	100.0	1,794
Paper products	221.4	355.0	23.82	25.25	87.4	1,584
Petroleum and coal products	617.6	109.7	117.88	61.22	97.4	830
Plastics and rubber products	297.4	704.0	40.24	74.86	90.0	944
Primary metals	303.4	363.9	154.95	210.51	82.7	1,150
Printing and related support activities	85.1	343.8	4.91	5.84	93.5	1,736
Textile mills and textile product mills	47.2	175.2	10.82	29.40	100.9	1,505

Subsector	Gross Output (\$B)	Employment (000s)	Exports (\$B)	Imports (\$B)	Labor Productivity (Index)	HHI
Wood products	172.9	403.9	7.90	23.55	93.6	1,962

Food and beverage/tobacco is the largest subsector by output and employment, with the lowest import-origin concentration (HHI = 837), consistent with a diversified agricultural input base. Wood Products and Printing show the highest concentration (above 1,700). Computer and electronic products shows the largest trade imbalance in absolute terms (\$302.7B exports vs. \$709.3B imports), consistent with documented U.S. dependence on foreign semiconductor and electronics manufacturing capacity (Chapter One).